

Aside on Experiments; Introduction to R & RStudio

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MATH 240

Lecture #3

Reminders

- Homework #0 is due **tonight** at 11:59pm
 - ▶ Includes survey, setup of tech, and turning in quick problem
- Lab #1 is due on **Tuesday** at 11:59pm via Gradescope
- Office hours: none on Monday; Tuesday from 3-4:30pm in Rocky 106A

Experiments vs. Observational Studies

- In lab settings, researchers may want to assign treatments to cases
 - ▶ This type of assignment is called an **experiment**
 - ▶ Used to show causal relationship between two variables
- In cases where we use no treatment, we are dealing with **observational data**
 - ▶ Used to show associations or form hypotheses

Explanatory Variable vs. Response

- We might care about the **relationship** between any two variables, in which:
 - ▶ x is the explanatory variable, and
 - ▶ y is the response
- Useful in experiments, in which we want to see if the explanatory variable has an effect on response
- Confounders might play a role
 - ▶ e.g. higher ice cream sales trigger higher swimming deaths
 - ▶ What is a confounding variable here?

Learning Objectives

- Familiarize yourself with the layout of RStudio, including the use of a Quarto document, the console, and other aspects
- Identify basic components of R syntax
- Code up basic calculations using R code

Welcome to R and RStudio!

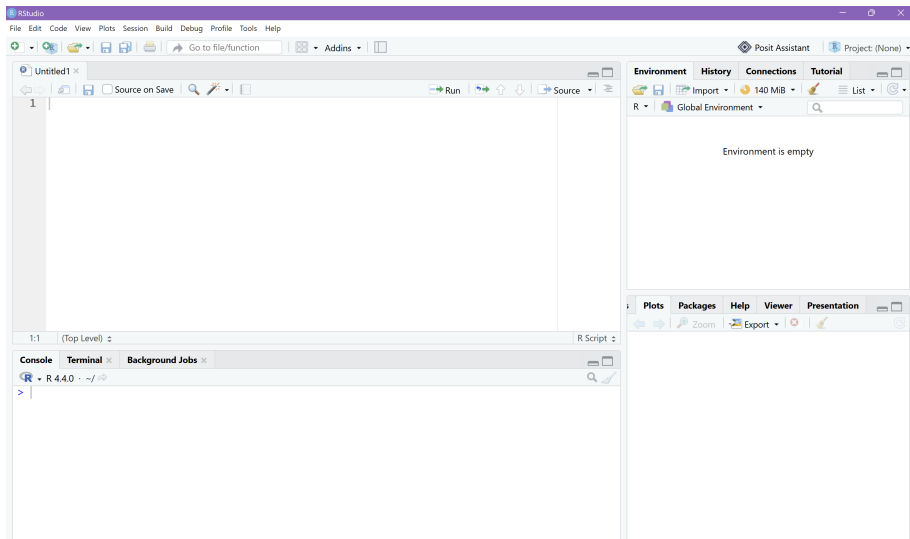


- **R:** Coding language used by statisticians and data scientists
 - ▶ Based primarily on C code
 - ▶ Contains syntax-friendly functionality
 - ★ Personal experience from already learning Python: easy transition, questionable syntax
- **RStudio:** Integrated development environment (IDE) for R
 - ▶ i.e. visual and user-friendly way to use R
 - ▶ Can run code, see results, and manage files all in one window
 - ▶ Personal experience: what I was taught, but Posit wants to move on to...

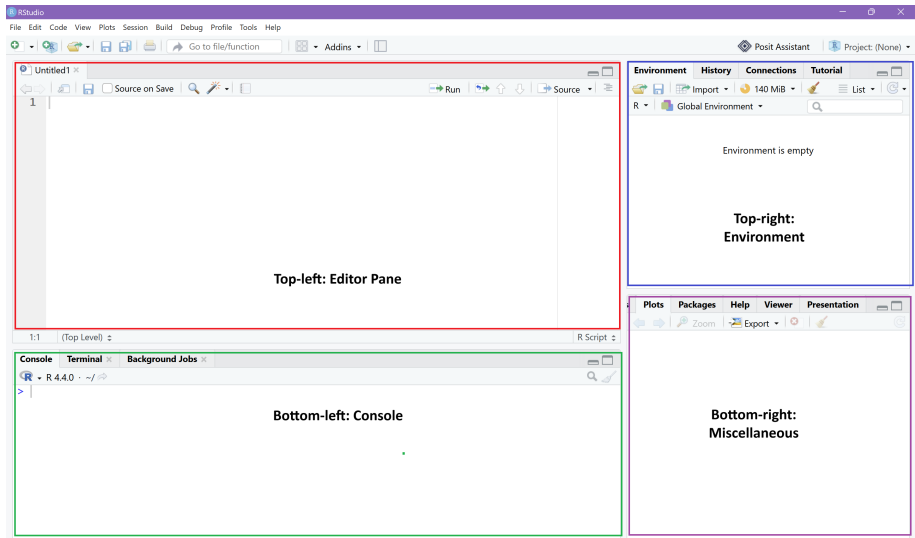
Installing R and RStudio

- Link for instructions
- Two-part process
 - ▶ Install R
 - ▶ Install RStudio

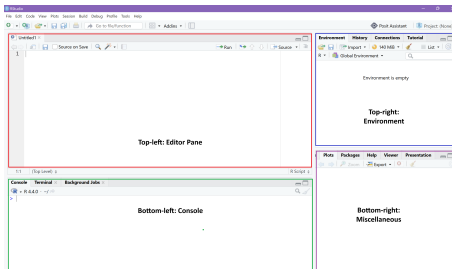
RStudio's User Interface



RStudio's User Interface

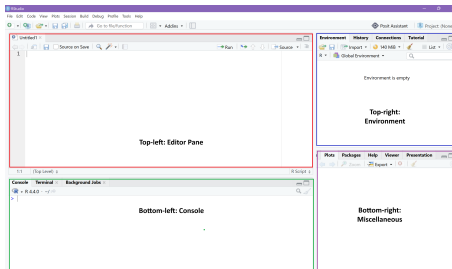


RStudio's User Interface



- **Editor:** Used for editing scripts, files, documents
 - ▶ Quarto documents (.qmd, combines code and prose)
- **Console:** Used for single-line commands; are not saved in a file
 - ▶ e.g. check if code runs; calculations

RStudio's User Interface



- **Environment:** Used to see which items are stored in memory (temporarily)
 - ▶ Data frames, vectors, variables
- **Miscellaneous:** Hosts many different tabs
 - ▶ Plots: to view plots you've run
 - ▶ Help: to find help files associated with functions
 - ▶ Packages: to see what packages you've installed

The Syntax (Console)

- Syntax consists of functions and objects
 - ▶ Object-oriented programming!
 - ▶ Functions: `function(input1, input2, ...)`
 - ▶ Objects: `x`, `data`

```
x <- c(1, 2, 3)
mean(x)

## [1] 2
```

Calculations (Console)

```
82 + 18
```

```
## [1] 100
```

```
82 - 18
```

```
## [1] 64
```

```
82 * 18
```

```
## [1] 1476
```

```
82 / 18
```

```
## [1] 4.555556
```

Note: Treat it like using a calculator (e.g. order of operations)

Vectors (Console)

- Building block for data storage in R
- One-dimensional
- Used for numerics, characters, and factors
- To build: `c()`

```
x <- c(1, 2, 3, 4)
```

Vectors (Console)

```
# Numeric  
x <- c(1, 2, 3, 4)  
x  
  
## [1] 1 2 3 4  
  
class(x)  
  
## [1] "numeric"
```

Vectors (Console)

```
# Character
y <- c("n", "e", "s", "w")
y

## [1] "n" "e" "s" "w"

class(y)

## [1] "character"
```


Vectors (Console)

```
# Factor
z <- factor(y)
z

## [1] n e s w
## Levels: e n s w

class(z)

## [1] "factor"
```

Data Frames (Console)

- Data frames are a series of vectors put together
 - ▶ Vectors make up the columns

```
data <- data.frame(x, y)
```

```
data
```

```
##      x y
```

```
## 1 1 n
```

```
## 2 2 e
```

```
## 3 3 s
```

```
## 4 4 w
```

Data Frames (Console)

- Accessing columns: `$`, `[, j]`
- Accessing rows: `[i, ]`
- Accessing cells: `[i, j]`

```
data <- data.frame(x, y)
```

```
data$x
```

```
## [1] 1 2 3 4
```

```
data[, 1]
```

```
## [1] 1 2 3 4
```

```
data[2, ]
```

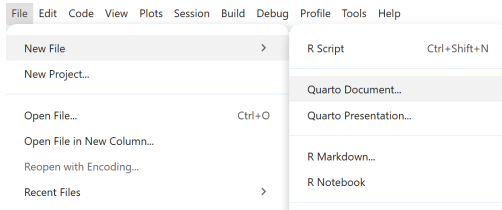
```
##    x y
```

```
## 2 2 e
```

```
data[2, 1]
```

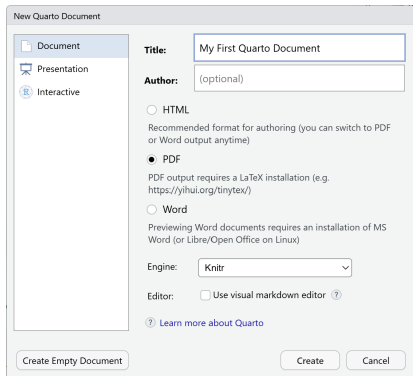
```
## [1] 2
```

Writing in a Quarto Document



- We'll be using Quarto documents (.qmd) to complete lab assignments
 - ▶ Integrates R code and written text

Writing in a Quarto Document



- File > New File > Quarto Document
- PDF option (we'll need to install the `tinytex` package)
- Use `install.packages("tinytex")` in console

Knitting a Quarto Document

Knitting a Quarto document will allow us to convert the prose and code (with its output) into a readable .pdf file.

- ➊ Write R code and text inside a Quarto document (in the editor pane)
- ➋ Click "Render" to view/publish the work as a pdf document
- ➌ Repeat steps 1 and 2 many times

Coding and Debugging

- Keep code/prose neat, organized, and reproducible
- Use comments to annotate code that explains what it does briefly
- AI is your friend here (in the old ages, Google/StackExchange)
 - ▶ e.g. I'm getting this error [here]. Explain why I'm getting this error.
 - ▶ Remember: treat AI as a second set of eyes, not your assistant

Getting Help

- Help functions and documentation
 - ▶ `?factor`
 - ▶ Search “factor” in Help tab
- Post R questions on Slack!

For Next Week!

- Homework #0 due tonight at 11:59pm via Gradescope!
- Lab #1 due Tuesday at 11:59pm via Gradescope
- First checkpoint: next Friday in class
- No class on Monday! Enjoy the holiday weekend :)
- On Wednesday: exploratory data analysis!